

Time: 3hours

Marks: 80

NB.

- 1) Question number 1 is compulsory
- 2) Attempt any three out of the remaining five questions.
- 3) Assume suitable data if necessary and justify the assumptions.
- 4) Figures to the right indicate full marks

Q1

Answer the following

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- a) Explain the significance of data science considering Volume and Dimensions of Data.
- b) Explain performance evaluation with respect to Time series forecasting
- c) Write a note on measure of spread
- d) Write a note on Applications of Data Science

Q2

- a) Explain Cross Validation, K-fold cross validation in detail
- b) Explain the Data Science Process in detail

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Q3

- a) What is Data Visualization? What is Multivariate data? Explain with an example Scatter plot.
- b) What are Outliers and their Causes? Can Statistics be used to detect Outliers if yes, Explain.

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Q4

- a) Explain SMOTE in detail
- b) Calculate the Bowley's coefficient of skewness from the following distribution.

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Income	NO. of persons
30 - 40	8
40 - 50	24
50 - 60	48
60 - 70	68
70 - 80	30
80 -90	13
90 - 100	9

Q5

- a) Explain Smoothing methods in Time Series Forecasting
- b) A stenographer claims that she can type at the rate of 120 words per minute. Can we reject her claim on the basis of 100 trails in which she demonstrates a mean of 116 words with a standard deviation of 15 words? Use 5 % level of significance, $Z_\alpha = 1.96$.

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Q6

- a) What are Recommendation engines? Explain.
- b) Time series analysis using linear regression

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